

BUSINESS ANALYTICS

Group Project Announcement

House Price Prediction for a Real Estate Listing Platform

An End-to-End Analytics Project: From Raw Data to a Monitored, Deployed Model

Course	Business Analytics
Project Type	Group Project (4–5 students per team)
Domain	Real Estate — Property Valuation
Deployment Target	Cloud free tier (e.g., Render, Hugging Face Spaces, Streamlit Community Cloud)
Coding Requirement	All code must be written by team members (no no-code/AutoML platforms)

1. Project Overview

Real estate listing platforms and buyers alike need a fast, defensible estimate of a property's fair market value — far ahead of a formal appraisal. An inaccurate estimate either scares away serious buyers with an inflated asking price or leaves money on the table for the seller. This project asks each team to act as the valuation-analytics function of a real estate platform and answer a concrete business question:

“Given a property’s characteristics and location, what is its estimated market price, and how confident is that estimate?”

Teams will build a regression model that predicts the sale price of a property from structural, locational, and condition-related features. The model will be engineered, deployed as a working application, and monitored after deployment — mirroring the full lifecycle of an analytics product in industry.

2. Data Sources

Teams must combine the following two data sources:

- **Kaggle dataset (required base):** “House Prices — Advanced Regression Techniques” — [kaggle.com/competitions/house-prices-advanced-regression-techniques](https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques). This provides structural attributes (area, number of rooms, age) and historical sale prices for homes in Ames, Iowa.
- **Synthetic data (required extension):** Each team must generate additional contextual data using Python (e.g., the Faker library plus custom business logic) — neighborhood amenity

scores (distance to school/hospital/transit), renovation history, days-on-market, and macro indicators such as a local interest-rate or price-index trend over time.

This combination ensures the project is not a simple “download-and-train” exercise: teams must justify the realism of the contextual features they generate and document the simulation logic rigorously.

Data Dictionary Requirement: Every synthetically generated field must be documented in a data dictionary specifying column name, data type, unit, valid range, and the generation logic or business assumption used.

3. Project Structure (8 Modules)

The project is organized into eight modules, suitable for an 8-week timeline (adjustable to fit the course schedule).

Module	Focus Area	Key Tasks
1	Business Understanding & Data Generation	- Define the business problem and key KPIs (prediction error tolerance, business cost of over/under-valuation) - Generate supplementary contextual property/neighborhood data in Python - Produce a complete data dictionary
2	Exploratory Data Analysis (EDA)	- Analyze distributions, outliers, and correlations between property attributes and sale price - Compare price patterns across location, property type, and age - Visualize relationships (e.g., price vs. area, price vs. distance to amenities) and check for multicollinearity
3	Data Cleaning	- Handle missing values, duplicates, and inconsistent categorical labels (e.g., property type naming) - Treat outliers (e.g., implausible area or price values, data-entry errors) - Document all cleaning decisions with before/after comparisons
4	Feature Engineering	- Construct derived features (price per square meter, property age, amenity-proximity score, renovation flag) - Encode categorical variables, scale numerical features, and address skewness in price (e.g., log transform) - Perform feature selection and check for multicollinearity (e.g., VIF)
5	Model Development	- Train and compare regressors: Linear/Ridge/Lasso Regression, Random Forest, XGBoost / LightGBM - Evaluate with regression metrics: RMSE, MAE, MAPE, and R^2; validate with k-fold cross-validation

Module	Focus Area	Key Tasks
		Analyze residuals to check model assumptions and identify systematic under/over-prediction
6	Model Deployment	- Package the model as an API (FastAPI or Flask) that returns a price estimate with a confidence range - Build a demo interface (Streamlit) simulating a property-valuation tool for agents or buyers - Deploy to a cloud free tier (Render, Hugging Face Spaces, or equivalent)
7	Model Monitoring	- Track data drift by comparing incoming listing data distributions to training data - Monitor rolling prediction error (RMSE/MAE) over time as new sales data arrives - Build a simple monitoring dashboard (e.g., using the open-source Evidently library) and define retraining triggers
8	Final Report & Presentation	- Translate findings into pricing-strategy recommendations and highlight key value drivers - Discuss limitations and future improvements - Present to the class with a live or recorded demo

4. Deliverables

Deliverable	Format
Source code (notebooks/scripts)	GitHub repository
Data dictionary	Excel or Markdown file
Deployed API + valuation tool demo	Live link (Render / Hugging Face Spaces / Streamlit Cloud)
Monitoring dashboard or report	Notebook or Evidently report
Final report and slides	PDF and PPTX

5. Suggested Team Roles (4–5 members)

Role	Responsibilities
Data Engineer	Synthetic contextual data generation, data dictionary, data cleaning
Data Analyst	Exploratory data analysis and visualization of price drivers

Role	Responsibilities
ML Engineer — Modeling	Feature engineering and regression model development
ML Engineer — Evaluation	Cross-validation, residual analysis, and error-metric interpretation
MLOps Lead	Deployment and monitoring (may be combined with another role for 4-person teams)

6. Evaluation Focus

Grading will emphasize the rigor and clarity of each stage, rather than model accuracy alone:

- Quality and realism of synthetic contextual data, supported by a complete data dictionary
- Depth of EDA and clarity of insights communicated through visualization
- Soundness of cleaning and feature engineering decisions, with clear documentation
- Correct use of cross-validation and appropriate regression metrics (RMSE, MAE, MAPE, R^2)
- A working, accessible deployment (live link required)
- A functioning monitoring component with defined drift/performance triggers
- Originality of code — all code must be authored by team members

7. Submission Logistics

Instructors should insert specific dates for each milestone below before distributing this announcement.

1. Team formation and topic confirmation — [insert date]
2. Module 1–3 checkpoint (data + EDA + cleaning) — [insert date]
3. Module 4–5 checkpoint (features + models) — [insert date]
4. Module 6–7 checkpoint (deployment + monitoring) — [insert date]
5. Final submission and presentation — [insert date]

Questions about scope, data sources, or deployment options should be directed to the course instructor.